VSD HW2

Instructor: Lih-Yih Chiou

TA: Tom

Date: 9/22/2025

Outline

- Introduction
- IEEE-754
- Testbench
- Homework

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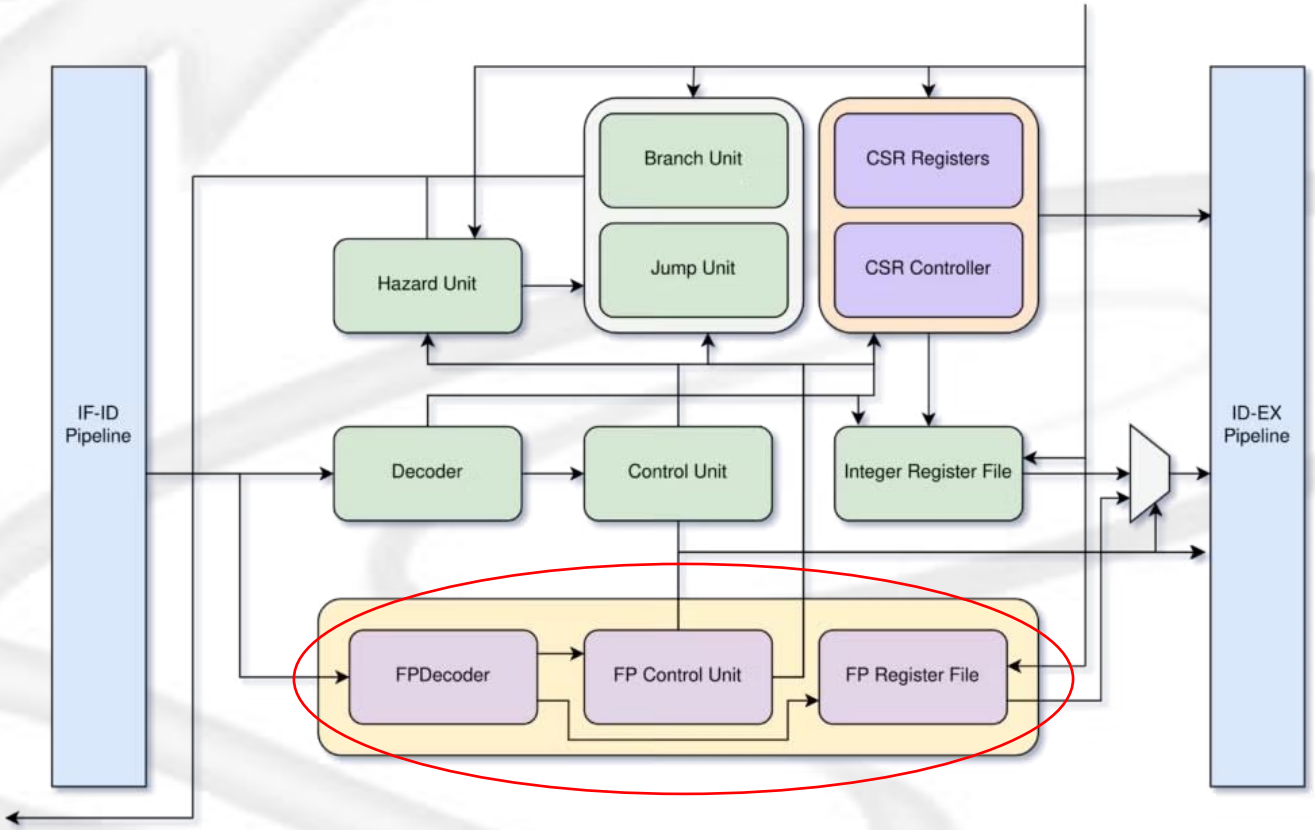
Introduction

- In order to design an IMF-CPU
  - ➔ We need an FPU.



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- Introduction
- IEEE-754
  - Floating Point Definition
  - Mantissa
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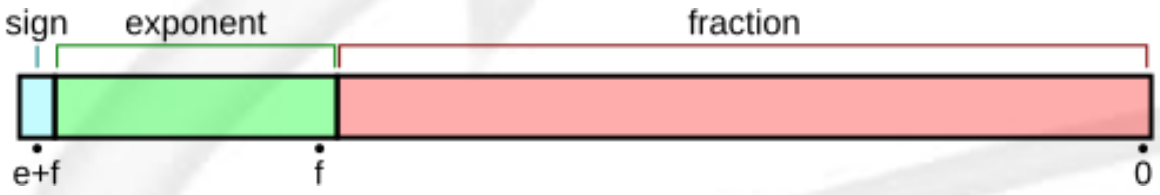
Flouting Point definition

- Single precision and double precision
  - Single precision(32bits) : 1(sign) + 8 (exponent) + 23 (frac)
  - Double precision(64 bits) : 1(sign) + 11(exponent) + 52 (frac)



Value = $(-1)^S * (1.F) * 2^{E-127}$

Data	0x40400000
S	0
E	(10000000)128
F	1.01(1.25)
Value	2.5

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Mantissa & GRS

- Mantissa(fraction part)
  - The normalized form is **1.F**
- GRS(Guard、Round、Sticky)
  - To align the exponents, we need to shift the mantissa.
  - G (Guard bit): the first bit shifted out of the mantissa.
  - R (Round bit): the second bit shifted out of the mantissa.
  - S (Sticky bit): if any bit after R is 1, then S is set to 1.

Fraction(1.F)	G|R|S
24bits	3bits

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Rounding Policy(round to even)

- Example

Round-to-even			
GRS	LSB	Action	Example
0xx	X	Discard	0.100_010 → 0.100
110 、101 、 111	X	Round up	0.100_110 → 0.111 0.100_101 → 0.111
100	1	Round up	0.101_100 → 0.110
100	0	Discard	0.100_100 →0. 100

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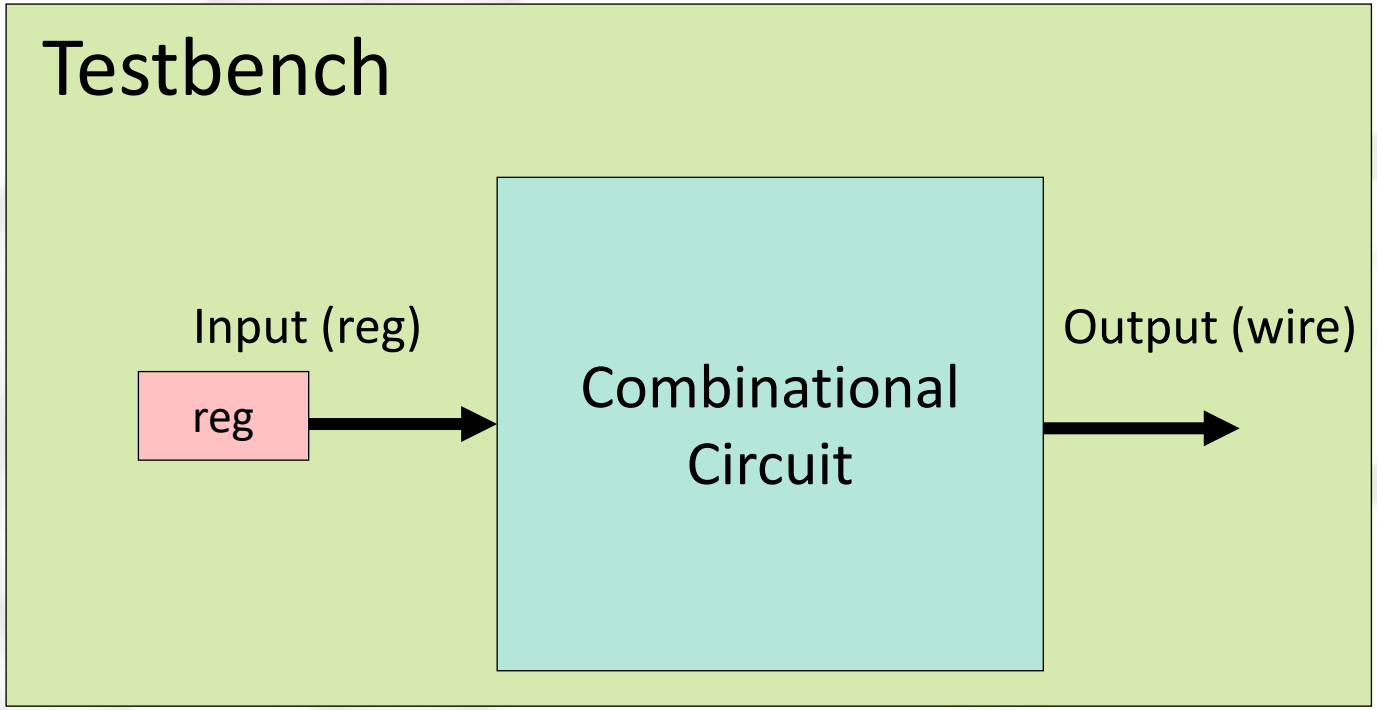
Testbench

- A testbench in Verilog is a separate module that is used to simulate and verify the functionality of your design.



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Testbench explained

```
module testbench;

    reg [3:0] patternA, patternB;
    reg [15:0] A_in , B_in;
    wire[15:0] Sum_o;
    wire       overflow;

    // your design
    RCA16 RCA(
            .a(A_in),
            .b(B_in),
            .sum(Sum_o),
            .overflow(overflow)
        );

    initial begin
        patternA = 4'd0; patternB = 4'd0;
        # (`CYCLE);

        for (i = 0; i < 4; i=i+1) begin
            for (j = 0; j < 4; j=j+1) begin

                // feed input
                A_in = {4{patternA}};
                B_in = {4{patternB}};
                # (`CYCLE/2);

                // check answer
                // .....
                # (`CYCLE/2);
                patternB = patternB + 1;
            end
            patternA = patternA + 1;
        end
    end
endmodule
```

Use 16-bit RCA as an example

The curly braces '{}' are used for concatenation and replication operations.

Assume patternA = 4'b0101,

A_in = { 4{patternA} } // replication

= {patternA, patternA, patternA, patternA} // concatenation

= 16'b0101_0101_0101_0101

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Prob.A (adder/sub)

- I/O information

Signal	Type	Bits	description
clk	Input	1	Clock
rst	Input	1	Reset signal
enable	Input	1	Enable signal
input_a	Input	32	First number
input_b	Input	32	Second number
sub	Input	1	0: add 1: sub
output_c	Output	32	Output answer
output_c_ready	Output	1	Output valid signal

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Add and Sub Steps (1/3) - Unpacking

- Unpack data to SEF and find special cases (NaN, Inf, Zero).

Type	Exponent	Fraction
NaN	255	Non-zero
Inf	255	Zero
Zero	0	zero

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Add and Sub Steps(2/3) – ADD/SUB

- Shift the smaller mantissa right, shift make exponents equal
  - ΔE = |Ea – Eb|
  - Right-shift the mantissa of the operand with the smaller exponent by ΔE
- Perform mantissa addition or subtraction depending on operand signs and the instruction. Below is special case

Operation	Result
Any nan in operand	Nan (32’hFFC00000)
Any inf in operand (except for sign mismatch)	Inf(32’hFF800000 or 32’h7F800000)
Sign mismatch inf (+inf + -inf)	Nan (32’hFFC00000)
0 + 0 (sign depand on & each other)	+0(32’h00000000) or -0(32’h80000000)
Only a 0 in operand	Return another one

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Add and Sub Steps(3/3) – Normalize && Round

- Handle carry (shift right, exponent +1) or cancellation (shift left, exponent −k).
- Apply rounding policy using GRS bits.
- Repack into 32bits data and output the result.

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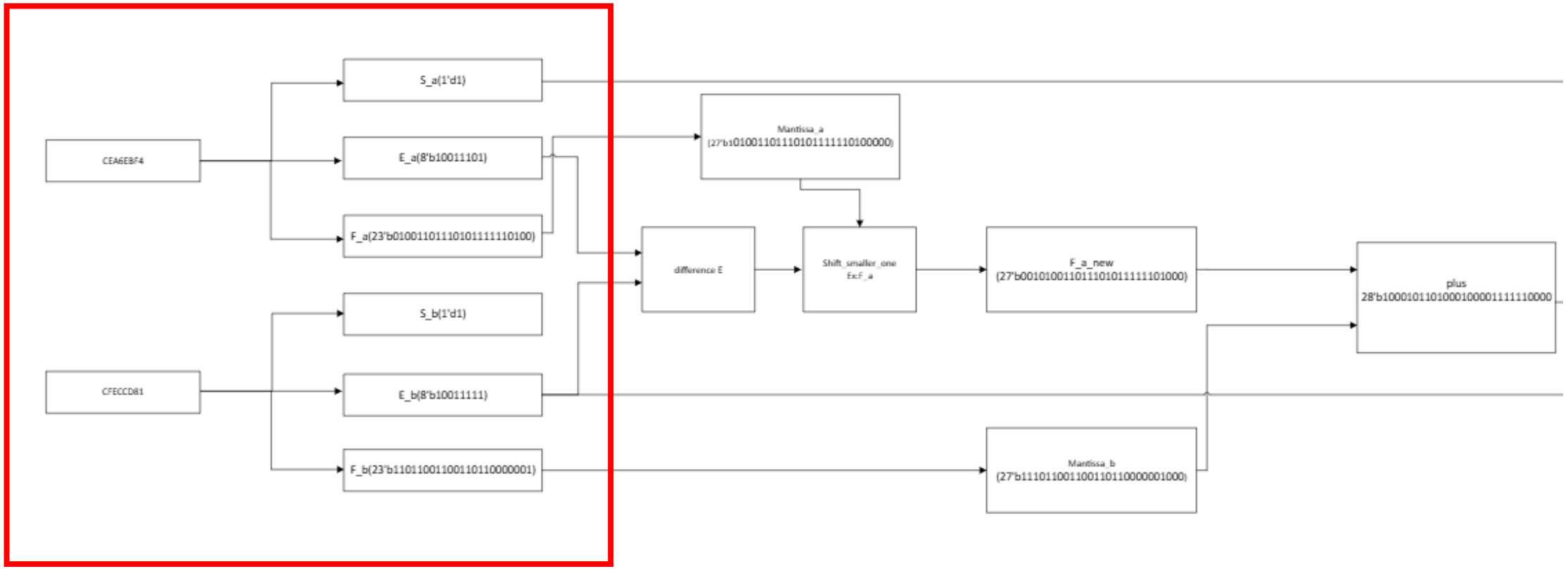
Example(1/3) - Unpacking



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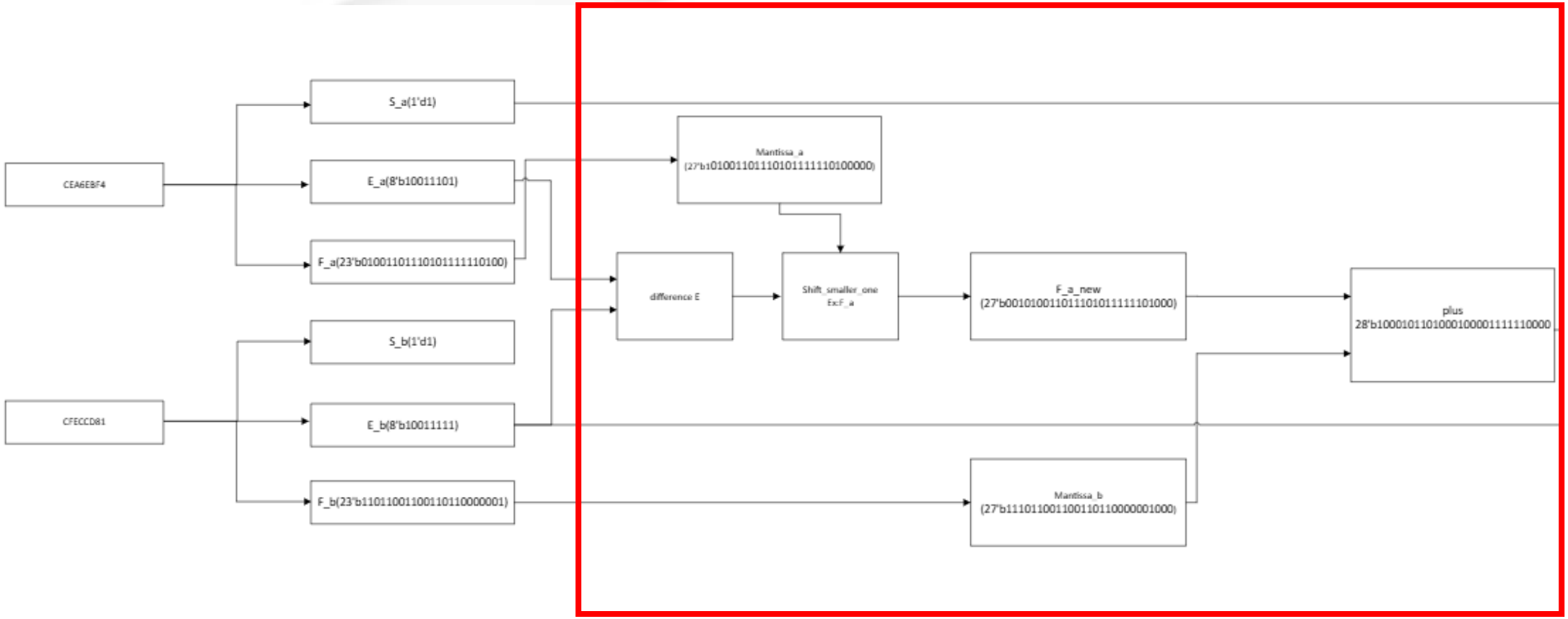
Example(2/3)– Add/Sub



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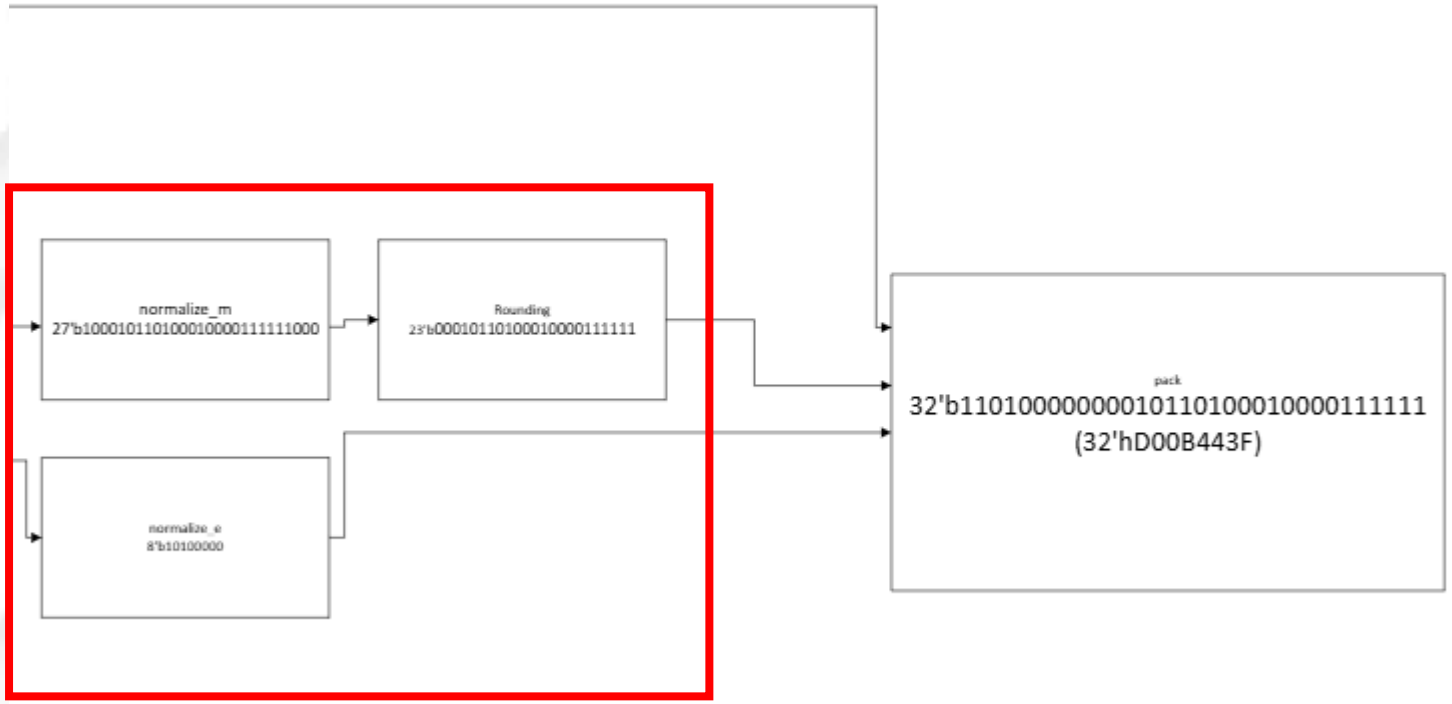
Example(3/3) – Normalize && Rounding



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Prob.B (comparator)

- I/O information

Signal	Type	Bits	Description
A	Input	32	First number
B	Input	32	Second number
mode	Input	1	Min : 0 Max : 1
answer	Output	32	Output answer

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Prob.B (comparator)

- Comparator Steps
  - Unpack to SEF
  - Check special case in adder chapter
  - Compare steps
    1. Sign bits
    2. Exponents
    3. Mantissas

Operation	Result
Any nan in operand	Nan (32’hFFC00000)
Both 0	32’d0

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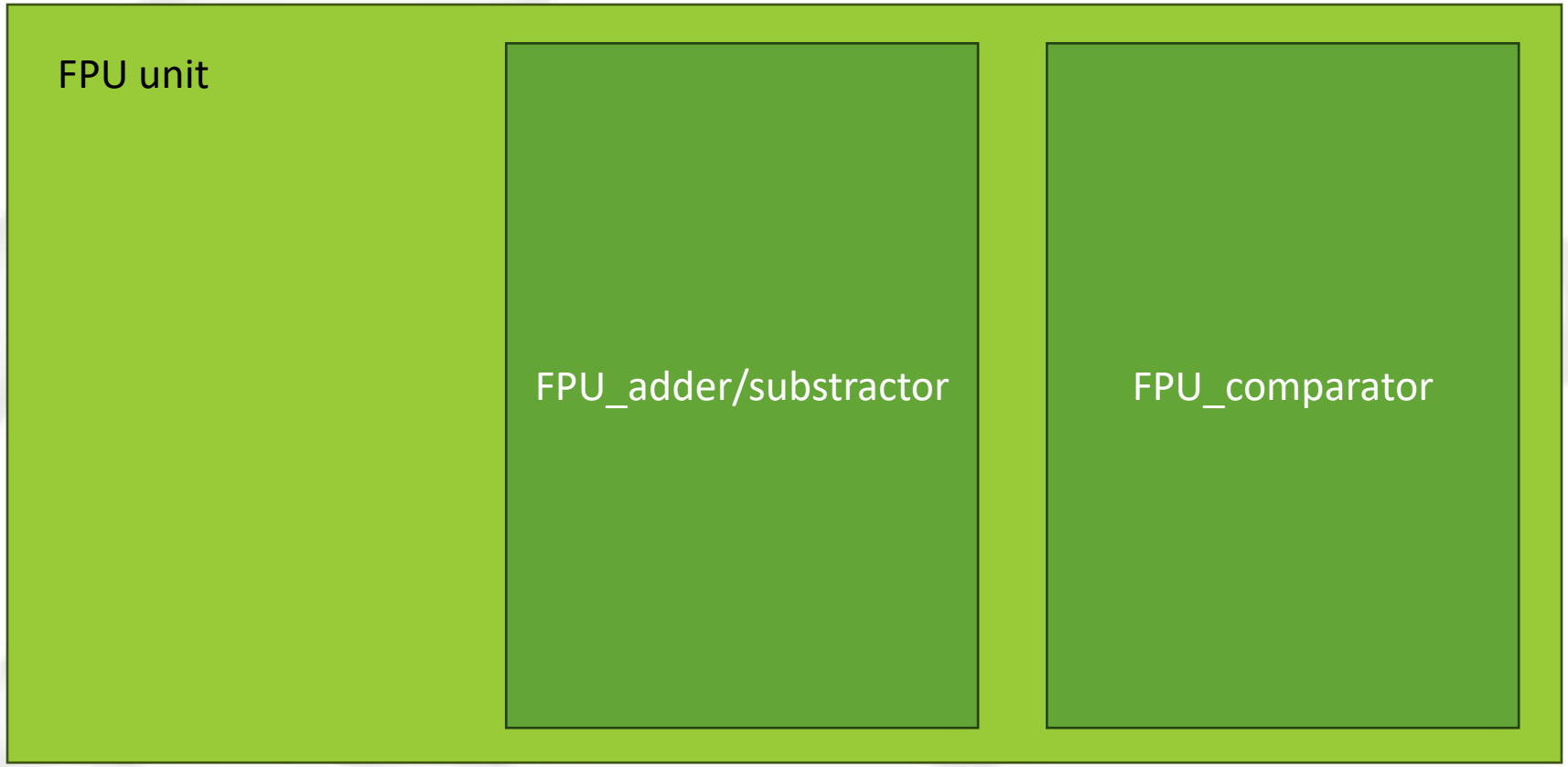
Prob.C (FPU unit)

- Block diagram



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Prob.C (FPU unit)

I/O define

Signal	Type	Bits	Description
clk	Input	1	Clock signal
rst	Input	1	Reset signal
enable	Input	1	Enable signal
ai	Input	32	First number
bi	Input	32	Second number
instruction	Input	2	0 : min 1 : max 2 : add 3 : sub
co	Output	32	Output answer
valid	Output	1	Output valid

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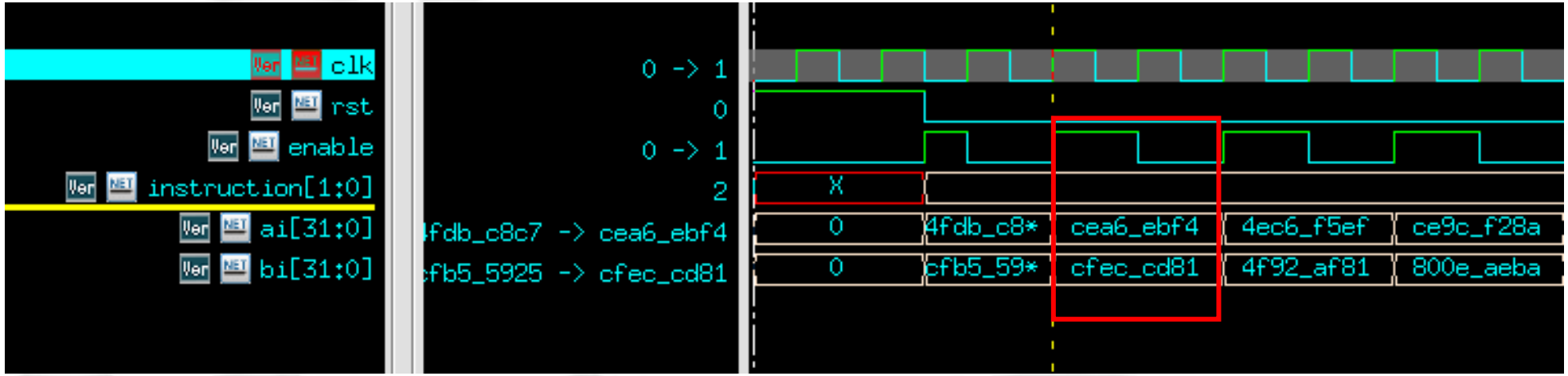
Data transfer(1/2)

- After reset low, pull high enable and feed data in!!



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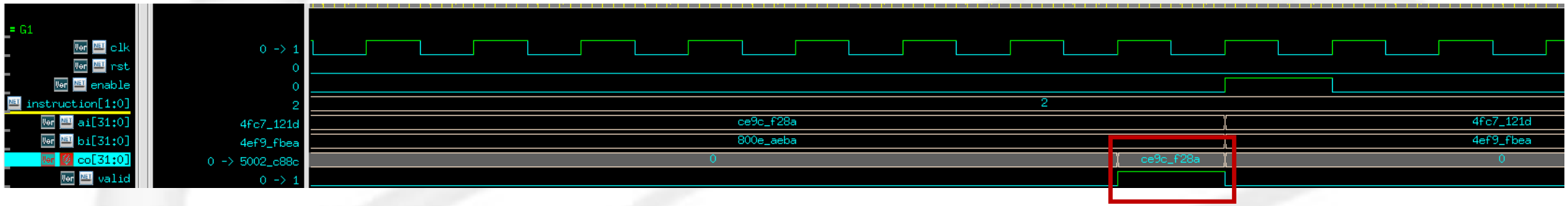
Data transfer(2/2)

- Wait valid signal pull high check the answer with golden, and next cycle new data feed into FPU.



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File hierarchy

- Homework submission must follow the file hierarchy!
  - StudentID_HW2.tar
    - StudentID_HW2
      - StudentID.pdf
      - Src
        - FPU.v
        - FPU_adder.v
        - FPU_comparator.v
      - Sim
        - FPU_tb.sv
        - tb_ex.sv
        - superlint.tcl
        - Pattern
          - pattern1.csv
          - pattern2.csv
          - pattern3.csv

Report must be in .pdf format

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Commands and Friendly reminder

- VCS Commands
  - Prob.A
    - vcs -R -full64 -sverilog FPU_tb.sv -debug_access+all +define+probA+FSDB
  - Prob.B
    - Vcs -R -full64 -sverilog FPU_tb.sv -debug_access+all +define+probB+FSDB
  - Prob.C
    - vcs -R -full64 -sverilog FPU_tb.sv -debug_access+all +define+probC+FSDB
  - Superlint
    - jg -superlint superlint.tcl

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Grading policy(1/2)

- Prob. A(35%) : FPU_add/sub
- Prob. B(25%) : FPU_comparator
- Prob. C(15%) : Pack A and B into FPU unit
- Testbench exercise(5%) : Complete the partially finished testbench provided by the TA.
- Report (20%)
  - The more complete you elaborate your design, the higher your grade is.
  - Do not copy and paste your whole code in the report, however, screenshot of some portion that help writing the report is permitted.
- Superlint coverage <= 80% (-10%)

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Grading policy(2/2)

- Submission that fails to follow the file hierarchy and format will result in a 10-credit reduction.
- Deadline: 2025/10/27 15:00
- Compress your homework in .tar format and upload it to Moodle.
  - Follow the file hierarchy for homework submission.
  - Ensure your code can be compiled and simulated in the SoC Lab environment.
  - Use *%tar -cvf StudentID_HW2.tar StudentID_HW2* to compress your folder. The compressed file should be named as "*StudentID_HW2.tar*"

    (e.g. *E24xxxxxx_HW2.tar*)

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Late Submission Policy

- Late submission within 24hr → score × 0.8
- Late submission within 48hr → score × 0.6
- Late submission within 72hr → score × 0.4
- Late submission more than 72hr → assignment will NOT be accepted. (score × 0)

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Thanks for Listening!

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